Building figures with Inkscape

Workshop Lead: Ariel Madrigal Aguirre

Registration link: NA

Approximate duration: 2 hours

Prerequisites:

1. Access to Inkscape latest version (free and open source)

Summary: (2-3 sentences summarizing the workshop)

This hands-on workshop will teach participants how to use Inkscape to transform a collection of independent plots into a polished, publication-ready figure. Participants will learn best practices for working with vector graphics, importing outputs from R and Python, arranging multi-panel layouts, and adding graphical elements such as arrows, annotations, and labels. By the end of the session, attendees will be able to create clear, professional figures suitable for manuscripts, posters, and presentations.

Learning Objectives: (List 2-5 learning objectives participants will learn upon completion of this workshop)

1. Distinguish between vector and raster graphics and identify when each format is appropriate for scientific figure preparation.
2. Combine plots generated in R, Python, or other software into a single figure using Inkscape.
3. Arrange and align multiple plots into a cohesive multi-panel figure using Inkscape's layout and alignment tools.
4. Customize figures with annotations, arrows, labels, and other graphical elements.
5. Export publication-ready figures in appropriate formats and resolutions for manuscripts, presentations, and posters.

Dataset:

Participants are encouraged to bring PDF figures generated in R or Python, either from previous workshops or their own projects. For those who do not have their own figures, a collection of mock RNA-seq analysis outputs will be provided to practice constructing a publication-quality multi-panel figure.

Content:

1. **Introduction to scientific figure design and Inkscape (25 min)**
    1. Why use Inkscape for scientific figures?
    2. Vector vs. raster graphics
    3. Common file formats (SVG, PDF, PNG, TIFF)

4. Best practices for publication-quality figures
5. Overview of the Inkscape interface and workflow

2. Importing and editing plots (30 min)

1. Understanding layers, groups, and objects
2. Importing images into Inkscape
3. Basic editing of imported graphics
4. Managing fonts and text elements

3. Break (5 min)

4. Building a multi-panel figure (45 min)

1. Alignment and distribution tools
2. Using guides, copying a color
3. Choosing fonts, colors, and line styles
4. Clipping and masking
5. Create a publication-style layout
6. Adding annotations (Drawing arrows and shapes)

5. Exporting and closing remarks (15 min)

1. Export options
2. Additional resources
3. Questions